

Data Analysis Report

Key Insights from SQL Queries

Order and Sales Analysis:

Overall Order and Revenue Performance -

- The platform processed a total of 15,000 orders, generating a cumulative revenue of approximately 3,808,899.
- The average order value stands at 253.93, indicating consistent transaction behavior across the customer base.

Order Status Distribution -

Order Status Total Orders Total Revenue

Pending	5,069	1,278,400.04
Delivered	5,057	1,284,616.01
Shipped	4,874	1,245,883.14

- These figures show that while the number of delivered orders is close to pending and shipped ones, a significant portion of the fleet is still transitioning or awaiting fulfillment.
- Collectively, 9,943 orders (66.3%) are still pending or shipped, which may impact revenue realization and customer experience.

Undelivered Orders -

Order Status Number of Orders

Pending	5,069
Shipped	4,874

- The high number of undelivered orders presents an opportunity to optimize the logistics and fulfillment pipeline.

Recommendations for Alt Mobility -

1. Optimize Fulfillment Efficiency

- Prioritize orders stuck in pending or shipped status.
- Review logistics workflows, vendor partnerships, and asset dispatch processes to reduce delivery delays.

2. Strengthen Retention Strategy

- Implement loyalty programs and follow-up campaigns to encourage repeat purchases.
- Offer personalized recommendations and discounts based on past behavior.

3. Segment and Target Customer Groups

- Use order frequency and value to identify high-potential customer segments.
- Tailor promotions for dormant, regular, and high-spending customers to increase conversion.

4. Monitor Payment and Order Completion Funnel

- Extend this analysis by investigating payment success/failure trends to detect and eliminate friction in the transaction process.

Customer Analysis:

Repeat vs. One-Time Customers

Customer Type	Number of Customers
One-time	2,932
Repeat	4,402

- A majority of customers (60%) have placed more than one order, indicating a strong base of repeat customers.
- Repeat buyers are likely more engaged and satisfied, contributing to a more stable revenue stream.

Revenue Contribution by Customer Type

Customer Type	Total Orders	Total Revenue
One-time	2,932	748,180.15
Repeat	12,068	3,060,719.04

- Though one-time customers make up 40% of the user base, they account for only 20% of revenue.

- In contrast, repeat customers contribute over 80% of the total revenue, emphasizing their importance to business growth and profitability.

Characteristics of Loyal Customers

An in-depth breakdown of repeat customers (placed >1 order):

- Customers like ID 2633, 9955, and 9476 have placed 7–8 orders each, spending over 1,700–2,000 individually.
- Average order values among loyal customers range from ₹138 to ₹396, indicating diverse purchasing behavior.

Key Insights -

- High Loyalty Potential: A large portion of the customer base exhibits repeat behavior, making them ideal targets for retention strategies.
- Revenue Dependence on Repeat Users: The business heavily depends on a relatively smaller group of loyal users for the majority of its revenue.
- Spending Patterns: Loyal customers do not necessarily have the highest individual order values, but their cumulative spending over time makes them highly valuable.

Recommendations -

Enhance Engagement for Repeat Customers -

- Launch loyalty programs or rewards to recognize and retain these high-value customers.
- Personalized offers or exclusive previews can further boost retention.

Convert One-Time Buyers -

- Follow up with one-time buyers through targeted campaigns and offers.
- Understand the reason for drop-off through feedback surveys or behavioral analysis.

Segmentation-Based Marketing -

- Segment customers based on frequency and spend.
- Tailor content and promotions to different segments (e.g., dormant users, regular buyers, high-spenders).

Payment Analysis:

Failed Payments with Shipped Orders -

Metric	Value
Total Orders	1,655
Total Order Amount	422,144.86
Average Order Value	255.07

Insight: Over 1,600 orders were shipped despite failed payments, implying a potential gap in payment validation workflows.

Failed Payments with Pending Orders -

Metric	Value
Total Orders	1,649
Total Order Amount	419,222.81
Average Order Value	254.23

Insight: These are orders that have not yet been fulfilled and may still be recoverable. A corrective action such as payment follow-up or order cancellation could reduce revenue risk.

Failed Payments with Delivered Orders -

Metric	Value
Total Orders	1,699
Total Order Amount	432,087.83
Average Order Value	254.32

Insight: This category poses the highest financial risk, as products are already delivered without successful payment.

Key Insights -

- A total of 5,003 orders have been processed across shipped, pending, or delivered statuses despite failed payments.
- Estimated potential revenue leakage from these failed transactions is over 1.27 million.

- All three categories show similar average order values (254–255), suggesting this issue is consistent across order sizes.

Recommendations -

- **Strengthen Pre-Dispatch Payment Validation**
Ensure payment status is verified before orders are marked as shipped or delivered.
- **Flag and Monitor Risk Orders**
Implement an alert system for orders with failed payments entering fulfillment.
- **Recovery Workflows**
For pending orders with failed payments, initiate automated recovery emails or payment retry mechanisms.

Orders details report:

Total Orders and Revenue -

Metric	Value
Total Orders	15,000
Total Order Value (₹)	3,808,899.19

Insight: Alt Mobility processed 15,000 orders, generating a total order value of approximately ₹3.81 million.

Successful Payments Received -

Metric	Value
Total Successful Payments	4,991
Total Amount Collected (₹)	1,257,085.15

Insight: Only about one-third of the total order value has been successfully collected via completed payments, highlighting a potential revenue collection gap.

Failed Payment Attempts -

Metric	Value
Total Failed Payments	5,003

Metric	Value
Attempted Amount	1,273,618.69

Insight: There were over 5,000 failed payment attempts, with potential revenue loss exceeding ₹1.27 million if unrectified.

Key Insights -

- The number of failed payments (5,003) is higher than successful ones (4,991), indicating a major issue in transaction processing or customer experience.
- Total paid (1.26M) is significantly lower than total order value (3.81M), suggesting a mismatch between order placement and payment fulfillment.

Recommendations -

- **Improve Payment Gateway Reliability**
Investigate technical or UX issues causing high payment failure rates.
- **Follow-up on Failed Payments**
Automate payment reminders or provide alternate payment options for failed transactions.
- **Align Order Fulfillment with Payment Confirmation**
Ensure strict control to prevent fulfillment without confirmed payments.

Customer Retention Visualization Explanation -

The customer retention visualization tracks how many customers placed repeat orders month-over-month. By extracting the month from each order's date and grouping orders by customer ID and order month, the analysis identifies customers who ordered two or more times. These repeat customers are then counted for each month.

This helps visualize retention by showing:

- Which months had more loyal customers
- Fluctuations in repeat buyer behavior over time
- The effectiveness of retention efforts by observing whether repeat ordering increases

The plot provides a clear trend of customer loyalty, which is crucial for understanding long-term business sustainability.

